BANQUET HALL BOOKING SYSTEM

Smart Venue Reservation & Event Management Platform

□ Phase 3 — Requirement Analysis	Customer Journey | Requirements | DFD | Tech Stack

3.1 Customer Journey Map

The customer journey maps the complete experience of a client from initial awareness to post-event feedback. It highlights key touchpoints, emotions, and opportunities for improvement at each stage.

□ Discover	□ Inquire	□ Compare	□ Book	□ Event Day	★ Feedback
Search for halls online or via referrals	Check availability, pricing & packages	Compare 2–3 halls on features & cost	Confirm booking & make payment	Coordinate with venue staff on-site	Rate the venue & leave a review
□ Overwhelmed by options	□ Frustrated by slow responses	□ Confusion from inconsistent info	□ Anxious about confirmation	□ Relieved if smooth handoff	□ Satisfied if expectations met

Pain Points at Each Stage

Stage	Key Pain Point / Opportunity
Discover	No centralized platform — users rely on Google, word-of-mouth, or physical visits.
Inquire	Phone tag and delayed email responses create uncertainty about availability.
Compare	Inconsistent information across venues makes comparison time-consuming.
Book	Manual paperwork and cash transactions lack transparency and documentation.
Event Day	Last-minute miscommunications due to lack of shared coordination tools.
Feedback	No structured channel for reviews, leading to unverified word-of-mouth only.

3.2 Solution Requirements

Based on the customer journey analysis and stakeholder interviews, the following functional and non-functional requirements were identified:

Functional Requirements

ID	Requirement Description
FR1	User Registration & Authentication — Secure login for clients, venue managers, and admins with role-based access control.
FR2	Hall Listing & Search — Advanced search with filters: location, capacity, price range, date availability, and amenities.
FR3	Real-time Availability Calendar — Dynamic booking calendar preventing double bookings with instant conflict detection.
FR4	Booking & Payment — End-to-end booking flow with online payment gateway integration (Razorpay/Stripe).
FR5	Notifications — Email/SMS confirmations, reminders, and status updates for both clients and venue managers.
FR6	Reviews & Ratings — Post-event client feedback system with rating aggregation and moderation.
FR7	Manager Dashboard — Booking management, revenue reports, and client communication tools.

Non-Functional Requirements

Requirement	Specification
Performance	Page load < 2 seconds; search API response < 1 second under normal load.
Scalability	Support 10,000+ concurrent users; horizontal scaling via cloud deployment.
Security	JWT authentication, bcrypt password hashing, HTTPS, SQL/XSS injection protection.
Availability	99.9% uptime SLA with automated failover and database backups every 6 hours.
Usability	Mobile-first responsive design; WCAG 2.1 accessibility compliance.

3.3 Data Flow Diagram (DFD)

The DFD illustrates how data moves between external entities, processes, and data stores within the system. The top-level (Level 0) context diagram is described below:

Level 0 — Context Diagram

Client (External Entity)	→	**Booking System (Central Process)**	→	**Venue Manager (External Entity)**

Data Flows: Client sends search queries, booking requests, and payment data → System processes requests, checks availability, and triggers notifications → Venue Manager receives booking alerts, confirms availability, and updates pricing.

Level 1 — Key Processes

P1	User Authentication — Validates credentials, manages sessions, and enforces role-based access.
P2	Hall Search & Browse — Processes search queries, applies filters, and returns sorted results.
P3	Availability Check — Queries booking calendar to confirm slot availability in real-time.
P4	Booking Management — Creates, updates, cancels bookings; assigns unique booking IDs.
P5	Payment Processing — Interfaces with Razorpay/Stripe; records transactions and receipts.
P6	Notification Engine — Triggers email/SMS for confirmations, reminders, and status changes.
P7	Review & Rating — Stores post-event ratings and aggregates venue scores.

3.4 Technology Stack

Frontend	Backend	Database

React.js / Next.js with Tailwind CSS. Responsive UI for web and mobile browsers.	Node.js with Express.js (REST API) or Django (Python) for business logic and API layer.	MongoDB (NoSQL) for flexible document storage or PostgreSQL for relational data.
Authentication	**Payment**	**Deployment**
JWT-based authentication with bcrypt password hashing and OAuth2 social login.	Razorpay / Stripe integration for secure online payment processing and receipts.	AWS EC2 / Vercel for frontend, Docker containers, Nginx reverse proxy.